

GramSetu AI (ग्रामसेतु AI)

Offline Rural AI Assistant & Multilingual Agent Orchestrator



Model: Qwen 2.5 (3B-Instruct)



Engine: Deterministic SQLite Memory + Verified RAG



Mode: 100% On-Device / Offline



UI: Web PWA & Flutter Mobile

SLIDE 1

1. Project Kya Hai? (What is GramSetu AI?)

Core Pitch: GramSetu AI ek voice-first, fully offline generative AI application hai jo rural Bharat aur village populations ke liye banaya gaya hai. Yeh bina kisi internet connection ke kheti (agriculture), sarkari yojana (govt welfare), primary health/first-aid aur daily problem solving me instantly guide karta hai.

Main Pillars:

- **100% Offline & Local:** Device ke andar chalne wala local AI engine (Qwen 2.5 3B via Ollama/llama.cpp).
- **Multilingual Voice Support:** Natural Hinglish, Shuddh Hindi (Devanagari) aur Punjabi (Gurmukhi) me baatcheet.
- **Zero Hallucination Guarantee:** Deterministic SQLite User Memory aur curated ground truth RAG engine se verified answers.
- **Specialized Agent Tools:** Acre-to-Bigha land conversion aur DAP/Urea/Seed dosage calculator.

SLIDE 2

2. Kyu Banaya? (Problem Statement & Rural Pain Points)

Real-World Rural Problem	Gramin / Kisan Ki Samasya	GramSetu AI Ka Exact Solution
1. Zero / Slow Internet	Khet me ya dur-daraz gaon me ChatGPT ya cloud apps bilkul nahi chalte.	100% Offline Local Inference — Device me hi process hota hai.
2. Language Barrier	Kisan English ya complicated robotic Hindi nahi samajh paate.	Natural Hinglish & Voice — Bolkar sawal pucho, desi bhasha me to-the-point jawab.
3. Galat AI Jawab (Hallucinations)	Cloud LLMs kheti aur dawaiyo me ulte-seedhe numbers bol dete hain.	Deterministic Memory & RAG — 100% verified sarkari aur scientific data.
4. Calculation Ki Kathinai	Zameen ka hisab (Acre, Bigha) aur khad ki matra nikalna mushkil hota hai.	Automated AgriMath Tools — 1 click me exact bags aur kg hisab.

3. Kaise Kaam Karta Hai? (System Architecture)

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USER INTERFACE (Web PWA & Flutter App) | | • Voice Input (Mic) • Hinglish/Hindi Chat •
Deep Think Panel | +-----+
-----+ | (User Query) ▼ +-----+
-----+ | 2. PREPROCESSING & ROUTER ENGINE | | • Language Detection (Input
vs Requested Script: Hindi/Hinglish/Punjabi) | | • Intent Classifier (Agri / Govt Schemes
/ Medical / Coding / Memory Query) | | • Tone Analyzer (Urgent / Casual / Technical /
Formal) | +-----+
--+ | _____ ▼ ▼ +-----+
-----+ +-----+ | 3. DETERMINISTIC MEMORY | | 4. AGENT TOOLS
& OFFLINE RAG | | • SQLite persistent facts | | • AgriMath (Acre → Seed & Khad) | | •
User Name, Land, Crop | | • Verified Knowledge (KCC, PMFBY) | | • Multi-turn pronoun parser
| | Fever First Aid, Arduino/IoT) | +-----+ +-----+
-----+ | | _____ | (Enriched
Grounded Context) ▼ +-----+
-----+ | 5. ON-DEVICE AI INFERENCE ENGINE | | • Qwen 2.5 (3B-Instruct) via Local
Ollama / Engine | | • Low Temp (0.15) + High Repeat Penalty (1.45) | | • Strictly To-The-
Point & Structured Bullet Reasoning | +-----+
-----+ | (Streamed Tokens) ▼ +-----+
-----+ | 6. OUTPUT SANITIZATION & DISPLAY | | •
Repetition Breaker & Dedup • Collapsible <think> • Local Audio TTS | +-----+
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User ka naam, zameen ka area aur context SQLite me store rehta hai. "Mera naam kya hai" jaise sawalo ka instant bina hallucination jawab milta hai.

KCC (Kisan Credit Card), PM Fasal Bima, PM-Kusum Solar Pump, Bukhar First Aid aur gehu/dhan ke certified facts instantly deliver hote hain.

 Kisan Mitra (Agricultural Advisory)

- **Gehu Ki Buwai:** 40-45 kg/acre seed rate, 1 bag DAP, 2 bags Urea.
- **Pehla Pani (CRI Stage):** Buwai ke 20-25 din baad zaroori sinchai.
- **Peela Ratua:** Propiconazole 25% EC @ 200ml/acre spray.

 Sarkari Yojana (Govt Welfare Schemes)

- **KCC Loan:** 4% effective interest rate, ₹1.6 lakh collateral free.
- **PM Fasal Bima (PMFBY):** 1.5% Rabi, 2% Kharif premium, 72-hr claim.
- **PM-KUSUM:** Solar agricultural pump par 60% tak subsidy.

 Swasthya & First-Aid Salah

- **Bukhar (Fever):** Paracetamol 500/650mg, cold sponging, ORS liquid.
- **Snake Bite Protocol:** Immobilize limb, DO NOT tie tourniquet, rush for Anti-Snake Venom (ASV).

 Upgraded AI Model: Qwen 2.5 (3B)

- 1.5B model se upgrade karke **3B parameters** model lagaya gaya hai.
- Strict **To-The-Point Reasoning** prompts: No rambling, no looping, crisp 2-4 points direct answer.

SLIDE 5

5. Kab Build Hua? (5-Day Project Execution Timeline)

Day / Milestone	Core Focus & Deliverables	Status
Day 1: Foundation	Architecture planning, problem discovery, SQLite schema design, project structure.	Completed
Day 2: AI Engine	Local inference backend, streaming HTTP server (server.py), Ollama bridge setup.	Completed
Day 3: Intelligence Layer	Deterministic Memory, Intent/Tone Router, Multi-turn context resolver, Offline RAG.	Completed
Day 4: UI/UX & Client	Responsive Web PWA, Flutter mobile UI, Deep Think reasoning panel, Voice STT/TTS.	Completed
Day 5: 3B Upgrade & Polish	Upgraded to Qwen 2.5 3B, anti-repetition guards, 10/10 automated test suite pass.	Completed

SLIDE 6 & 7

6. Demo Highlights & Presentation Pitch Tips

Live Demo Script (Presentation Me Kaise Dikhavein):

- Memory Step:** Type karein: *"Yash naam hai mera aur mere paas 4 acre zameen hai"* → System turant save kar leta hai.
- Verification:** Puchiye: *"Mera naam kya hai aur meri zameen kitni hai?"* → Instant zero-hallucination exact answer.
- Agri Query:** Puchiye: *"Gehu me pehla pani kab dena chahiye?"* → 20-25 din CRI stage ka crisp to-the-point guidance.
- Scheme Query:** Puchiye: *"Kisan credit card kya hota hai aur byaj kitna lagta hai?"* → 4% interest rate & application details.

Pitching Tips for Students (Presentation me bolne ke liye):

- **Start Strong:** *"Namaste sabhi ko. Aaj hum present kar rahe hain GramSetu AI — Bharat ke gaon aur kisano ke liye pehla 100% offline, voice-first multilingual AI assistant."*
- **Live Impact:** *"Is application ko chalane ke liye 1 byte internet ki zaroorat nahi hai. Khet me ho ya jungle me, AI aapki bhasha me madad ke liye hamesha tayyar hai."*
- **Closing:** *"GramSetu AI bridges the digital divide between urban tech and rural empowerment. Dhanyawad!"*